ASSIGNMENT 4

Instructions for Students

- Create separate .js files for each question.
- Use descriptive variable names, indentation, and comments to explain logic.
- Use only core JavaScript (ES6+) — no frameworks.
- Test using Node.js console or browser console.

Q 1: E-Commerce Product Manager (Classes + Objects)

You're building an **E-commerce Admin Panel**. Create a **`Product`** class with properties like `id`, **`name`, `price`,** and **`category`**.

- Add methods to:
    - Apply a discount (e.g., 10%)
    - Display product details in a formatted string
- Create multiple product objects and store them in an array.
- Display products with price > 1000 in the console using a filter.

Q 2: Student Form Validator (Forms + RegExp)

Create a **student registration form** with fields: **`Name`, `Email`, Phone, `Password`**.
Use **JavaScript + RegExp** to validate:

- Name: only alphabets
- Email: valid format (**`example@domain.com`**)
- Phone: exactly 10 digits
- Password: must contain 1 uppercase, 1 number, and 1 special character

Show red border + error message if invalid, green if valid.

Q 3: Library Management System (Classes + Objects)

Design a **`Book`** class with properties like **`title`**, **`author`**, **`ISBN`**, **`isIssued`**.

- Create methods **`issueBook()`** and **`returnBook()`**.
- Create an array of book objects.
- Display all available books (not issued).
- Allow a user to issue a book by searching ISBN.

Q 4: Custom Form Builder (Forms + Classes)

Build a dynamic **form generator** using a **`FormBuilder`** class that takes an array of field objects (e.g., **`{type: 'text', label: 'Username'}`**) and creates a form dynamically using **`innerHTML`**.

Note

- Add a method **`getFormData()`** that returns all entered values as an object when user clicks "Submit".

Q 5: Movie Ticket Booking (Objects + RegExp)

Build a movie ticket booking form for an app.
Validate user input:

- **`Name`** (alphabets only)
- **`Email`** (proper format)

- **`Seats`** (1 to 10 only)

After successful validation, store booking info in an object **`{ name, email, seats }`** and display the ticket details.

Q 6: Employee Management System (Classes + Object Methods)

Create a class **`Employee`** with properties: **`id`, `name`, `department`, `salary`.**

- Add method **`getAnnualSalary()`** and **`applyBonus(percent)`.**
- Create 5 employee objects and calculate annual salary for each.
- Use **`reduce()`** to calculate total annual payout of the company.

Q 7: Login Form Validation using RegExp

Create a login form with username & password.
Use RegExp to check:

- Username: at least 5 characters
- Password: at least 8 characters, must include number, uppercase, lowercase, special character

If validation passes → show success message; else → show specific error messages.

Q 8: Dynamic Object Updater

Create an object **`user = { name: "John", email: "john@mail.com", age: 21 }`**.
Build a **form** that allows editing these values.
On submit, the form should **update the user object** in real time.
Show updated object details below the form using DOM.

Q 9: Shopping Cart Total (Classes + RegExp for Coupon)

Build a shopping cart system using a `Cart` class.

- Add items with **`name`**, **`price`**, **`quantity`**.
- Calculate total using **`getTotal()`**.
- Apply coupon validation using RegExp:
    - Coupon format must be like **`SAVE20`** or **`DISC10`**.
- If valid, apply corresponding % discount and show final total.
